Binaural Spatial-Cue Guided Speech Separation and Spatial Re-Mixing in MATLAB

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GitHub project: https://github.com/shiyingh114-jpg/elec5305-project-540662535

Demonstration video:https://github.com/shiyingh114-jpg/elec5305-project-540662535/blob/main/demonstration%20video.mp4

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1 Introduction

Target speaker separation from competing speakers is a common task in the field of communication, hearing assistive technology, and spatial audio. Signals acquired using microphone arrays are a superposition of several sources, while a listener usually focuses on one source only. Separation of overlapping speech is especially complicated due to the similarity of frequency ranges and fast temporal changes of both target and interfering sources. The proposed separation method should preserve the target without introducing too much distortion. Within the binaural model, it is additionally needed to consider the inter-ear relation of signals, since changes in this relation might change perceived spatial properties of the scene.

Interaural level difference (ILD) and interaural phase difference (IPD) are two components of the binaural relation. Frequency dependent patterns of ILD and IPD could be different for sound sources with different directions. In case when a short time-frequency region is dominated by a single source, observed interaural relations in this region might help to distinguish this region from another source. It gives us additional information apart from just the mixture spectrum. The involved signals are the recordings at the left and right ears: any stereo channels don't necessarily have the same physical interpretation. HRIR measurements provide a controlled way to create ear signals with specific directions of the sources and corresponding clean references.

As a first option, we consider the spectral Wiener mask. It estimates the powers of the target and interference and attenuates the regions assigned to the interference. However, in case when the second speaker is an interferer, noise power estimation in a spectral Wiener mask could be wrong, which leads either to attenuation of target signal or preserving of interfering speech in the output signal. The spatial masking technique solves a different task – it compares observed binaural cues with the expected ones for the target. The efficiency of this technique depends on the ability of interaural cues to distinguish two sources, which makes angular separation an appropriate experimental parameter [1,2].

The core research question is: By how much can the use of binaural ILD/IPD cues improve soft-mask speech separation from a single conventional spectral mask, and how does the improvement depend on the angle of separation between the target and the interferer? The direction of the target is known. For the experimental setup, the target is kept in the frontal position relative to the listener, while the interferer's direction and input signal-to-noise ratio (SNR) are changed. Since there exist dry signals and rendering filters, any estimated output is verifiable against the corresponding binaural target.

In the current study, we compare ILD-only, IPD-only and combined spatial masks with a simple Wiener filtering and with the product of a spectral and spatial mask. We modify the product with the requirement for the lower bound on the Wiener gain. The modification takes into account the fact that the simple product might not outperform a single spatial mask since the spectral mask might attenuate a region that the spatial mask keeps. The modified rule puts some constraint on the additional attenuation. Its effect is investigated on the same mixtures since saving more energy for the target may mean preserving more energy for the interferer.

This project evaluates separation performance and spatial structure preservation simultaneously. Reconstruction performance is measured using reconstruction SNR and the scale-invariant signal-to-distortion ratio (SI-SDR), and reference-weighted ILD/IPD error values are used to characterize the preserved structure. The reconstructed target and the residual are further rendered in novel directions in order to demonstrate one of the practical applications. The current work is mostly devoted to the analysis of the effect of spatial cues and the effect of the modification of the product of masks.

2 Literature Review

2.1 Spatial separation

Mandel, Weiss, and Ellis define the interaural features within MESSL and solve for the sources using the expectation-maximization algorithm [1]. In their model, the source models are linked to probabilistic time-frequency masks, and assume that there is enough dominance in the sources in the particular regions of the spectrogram. Our method relies on the idea of cue matching, however, the target direction is supplied to the algorithm and constructs its template from the empirical head-related impulse response (HRIR). We neither estimate the mixture of the source models nor carry out the whole MESSL algorithm. The fixed target compatibility score makes the result more understandable, although it loses adaptiveness of source priors and competition between source models.

In Alinaghi et al., binaural models are combined with the information of the mixing vector [2]. As shown in their analysis, cue sources can have different informative strength in different acoustic environments, even when the source directions are close to each other. Therefore, the motivation for performing ILD/IPD ablation and angular sweep follows. Their joint model is more elaborate than the Gaussian mask used here; thus, the performance of their model cannot be used as a baseline number.

2.2 Cue preservation

The work by Van den Bogaert et al [3]. focuses on the multi-channel Wiener methods in the realm of binaural hearing aid devices, specifically concerning the reduction of noise in relation to the retention of the acoustic scene. Koutrouvelis et al [4]. make the connection between the spatial retention and the audibility of the sources after processing. Together, these papers show that suppression is not enough. They support the use of both waveform and spatial metrics in combination, although the scalar shared-ear mask used in this paper is much simpler compared to the multi-channel Wiener method.

2.3 Rendering and learned methods

The measured HRIRs for binaural rendering of Sadie II [5] together with the SOFA format for filters storage and geometry [6] allow controlling of source direction without any head model construction. It is useful for the separation experiments since original ear signals could be used. Differentiation of the HRIRs among individuals indicates that the result, achieved on one listener, does not apply to the behavior of a universal binaural system.

The learned models could incorporate speech structure beyond the restrictions of one time-frequency bin. In particular, Han et al. expand time-domain separation to binaural processing [9]. Wang et al. propose a light-weight complex network that solves the enhancement with maintaining spatial information using relative transfer function prediction [10]. Yang et al. investigate angular region separation based on the neural network which is trained on simulated microphone responses [11]; it should be mentioned that sensor configuration in this work is different from the measurements with ear HRIRs used in this paper. The comprehensive review by Wang and Chen includes discussion of the targets of training and generalization aspects as key design aspects of the supervised separation task [12].

These papers provide sufficient background for a simple classical experiment with clearly formulated assumptions. Known target template excludes the localization estimation, and separate masks emphasize the contribution of each cue. The limitations are obvious as well: learned speech model, adaptive two-source posterior distribution, and robustness to the room reverberation are not used. The lower bound of the Wiener factor serves as a simple increment of this approach. Comparison of the proposed algorithm to MESSL and neural network algorithms should be done by running them on the same mixtures and evaluation intervals.

3 Methods

3.1 Data and rendering

Three pairs of LibriSpeech dev-clean mono recordings [7] are chosen as target and interferers. These signals are re-sampled to 48 kHz and cut to their overlapping part, with an eight-second limit applied; no loopings of the signals shorter than the eight-second limit are used. The target excerpts have lengths of 5.855 s, 2.260 s, and 8.000 s. Dry signals are normalized to unit RMS. The rendering is done using the 48 kHz, 256-taps SADIE II H10 HRIR measurements in SOFA format [5,6]. The target is placed at azimuth 0° and elevation 0°. For the interferers specified at 15°, 30°, 60°, and 90°, the closest recorded angles are taken, that is, 13.3°, 30°, 60°, and 90° correspondingly.

$$x_e[n] = (s * h_{e,t})[n] + g\,(v * h_{e,i})[n], \quad e \in \{L, R\} \quad (1)$$

Let s and v be the sources dry, h be the corresponding ear-specific HRIR, and g specify the SNR of the input signal as −5, 0, or +5 dB. Linear convolution includes the filter tail. The gain is computed from the sum of the energy of both ears over the scoring period, but the interaural gain and delay measurements are kept intact. The clean rendered parts are saved as evaluation references. The target-silent prefix of 0.6 seconds is made available in order to provide a calibration section of only the interference.

3.2 Analysis and spatial masks

The STFT is obtained using the square root of periodic Hann window of size 1536 samples (or 32ms) along with hop size of 768 samples and 2048 point FFT in a sampling frequency of 48 kHz. The WOLA process and end padding are used for preservation of reconstruction limits. The evaluation does not consider the prefix, the number of samples of the target head-related impulse response (HRIR), and the analysis window, and hence the time values become 5.823 s, 2.228 s, and 7.968 s respectively for the three pairs. The STFTs of the left/right mixture provide the ILD and IPD while the HRIR template of the target direction gives the target values.

$$\mathrm{ILD} = 20\log_{10}(|X_L|/|X_R|), \quad \mathrm{IPD} = \arg(X_L X_R^*) \quad (2)$$

$$M_{\mathrm{sp}} = \exp\left(-\frac{\Delta\mathrm{ILD}^2}{2(4\,\mathrm{dB})^2} - \frac{\Delta\phi^2}{2(0.5\,\mathrm{rad})^2}\right) \quad (3)$$

The phase error is modeled as the wrapped phase function $\Delta\phi = \arg\{\exp[\mathrm{j}(\mathrm{IPD} - \mathrm{IPD_target})]\}$, which prevents the discontinuity around $\pm\pi$. The masks based either on interaural level difference (ILD) or on interaural phase difference (IPD) retain the respective exponential part from equation (3). The spatial weights are set to be neutral outside the 100–8000 Hz range, as well as for target-HRIR nulls that are lower than −60 dB. Mixture bins with power levels below −80 dB are silenced. The compatibility weights are not normalized to posterior probabilities.

3.3 Wiener fusion

The smoothing time constant for the average power of the mixture signal is set to 50 ms. The interference power is initialized using uncontaminated prefix frames and thereafter updated using a time constant of 0.3 seconds only when the energy of the mixture signal becomes less than 1.5 times the interference estimate.

$$W = \frac{\max(\widehat{P}_x - \widehat{P}_v, 0)}{\max(\widehat{P}_x - \widehat{P}_v, 0) + \widehat{P}_v}, \quad M_0 = M_{\mathrm{sp}} W \quad (4)$$

$$M_\lambda = M_{\mathrm{sp}}[\lambda + (1-\lambda)W], \quad \lambda = 0.5 \quad (5)$$

The constraint mentioned in equation (5) limits the effect of the Wiener term on suppression as long as spatial compatibility of the region remains valid. The effective suppression term is limited to [0.5, 1], and thus an affine transformation is employed here as opposed to clamping of W to 0.5. This manipulation results in at most 6.02 dB more attenuation compared to the case when the spatial gain is nonzero. With λ equal to zero, we get back the original term; for λ = 1, masking becomes spatially selective. 0.5 was decided on before conducting the extended SNR study. There is a single gain employed equally for both ears.

3.4 Evaluation

Three pairs × four angles × three SNRs give 36 combinations of scenes. In each scene, there are 8 values calculated from the mixture, the Wiener filter, the ILD, the IPD, the joint spatial, original product, the floored product, and the oracle power ratio, giving rise to 288 rows of scores. The oracle makes use of the clean target and interferer power, but the mask used in practice depends only on the mixture, target HRIR, and calibration data.

$$\mathrm{SNR}_{\mathrm{rec}} = 10\log_{10}\frac{\|S\|^2}{\|Y-S\|^2}, \quad \mathrm{SI{-}SDR} = 10\log_{10}\frac{\|\alpha S\|^2}{\|Y-\alpha S\|^2} \quad (6)$$

Reconstruction SNR compares the reconstructed output Y with the clean target S without gain fitting or alignment techniques. The scale-invariant signal-to-distortion ratio (SI-SDR) uses demeaned signals, where α is the inner product of the demeaned signals divided by the energy of the target [8]. Scores averaged across ears are reported. Improvement is calculated by subtracting the unprocessed mixture score from the corresponding output score. The frozen masks are separately applied to each known component resulting in the scores of target retention, interference retention, and output SIR. Retention is calculated as the ratio of the passed energy to the energy of the original component in dB; 0 dB denotes no energy loss.

The spatial error metrics are the target-energy-weighted ILD and IPD RMSE evaluated over the active reference frequency bins in the 100–8000 Hz range after waveform resynthesis and reanalysis. Cue-energy coverage determines the percentage of reference energy that results in output cues. Diagnostic analysis was split into low- and high-band parts at 1.5 kHz, but not in a way that would be considered a band-specific masking ablation. Short-term objective intelligibility (STOI) and listener ratings were not computed. In tables, arithmetic means across pairs are reported; all error bars in the performance plots are based on the sample SD. Dependent observations are scenes sharing utterances.

3.5 Re-mixing and verification

The setup uses pair 1 at 60° and 0 dB. Each recovered binaural signal is folded down to mono and rendered anew at -30° and +60° for the target and residual, respectively. Clean components are similarly treated through fold-down and new HRIR filtering in order to create corresponding references. This approach keeps the original HRIR coloration intact and does not involve a dry-source reconstruction. Validation in MATLAB shows overlapping-add behavior, finite outputs, target and residual reconstruction, with the highest complement error amounting to 7.11×10^{-15}. The existing 0 dB benchmark results remain unaltered; the setup, source hashes, and metrics are included in the code.

The setup goes through the exact same steps in any scene: loading and verifying the source files and SOFA geometry; rendering the two binaural components; setting the interferer gain; analyzing the mixture; estimating all masks; synthesizing the outputs; and scoring them in a unified way. The log records the requested and actual angles along with the rows in the metrics table. Ear normalization is not performed on the exported audio files; there is a common playback gain applied to both ears across all methods. This practice ensures that there will be no audible binaural and loudness inconsistencies introduced by different file exports.

There are three relevant technical subtleties for the interpretation of the experiment. First, using a larger FFT leads to denser frequency sampling rate but does not change the underlying resolution set by the window. Second, phase difference is circular and frequency-dependent, hence simple unwrapped subtraction would not work for cues' matching. Finally, scalar masking and inverse-STFT reconstruction of the output signal do not guarantee the preservation of all the coefficient values that were modified during synthesis. These factors justify using the wrapped phase model, reconstruction checking, and spatial evaluation in the waveform domain. The fixed tolerances and fusion weight make the ablation interpretable.

4 Results

Results initially highlight the contribution of the spatial cues, with a further analysis of the Wiener correction and the cost incurred in terms of interference preserved. Unless otherwise stated, all numbers refer to the mean values of the three speech pair conditions. Angles are based on the actual angles obtained, with the first one being 13.3˚, even though it was asked for at 15˚.

4.1 Angular separation

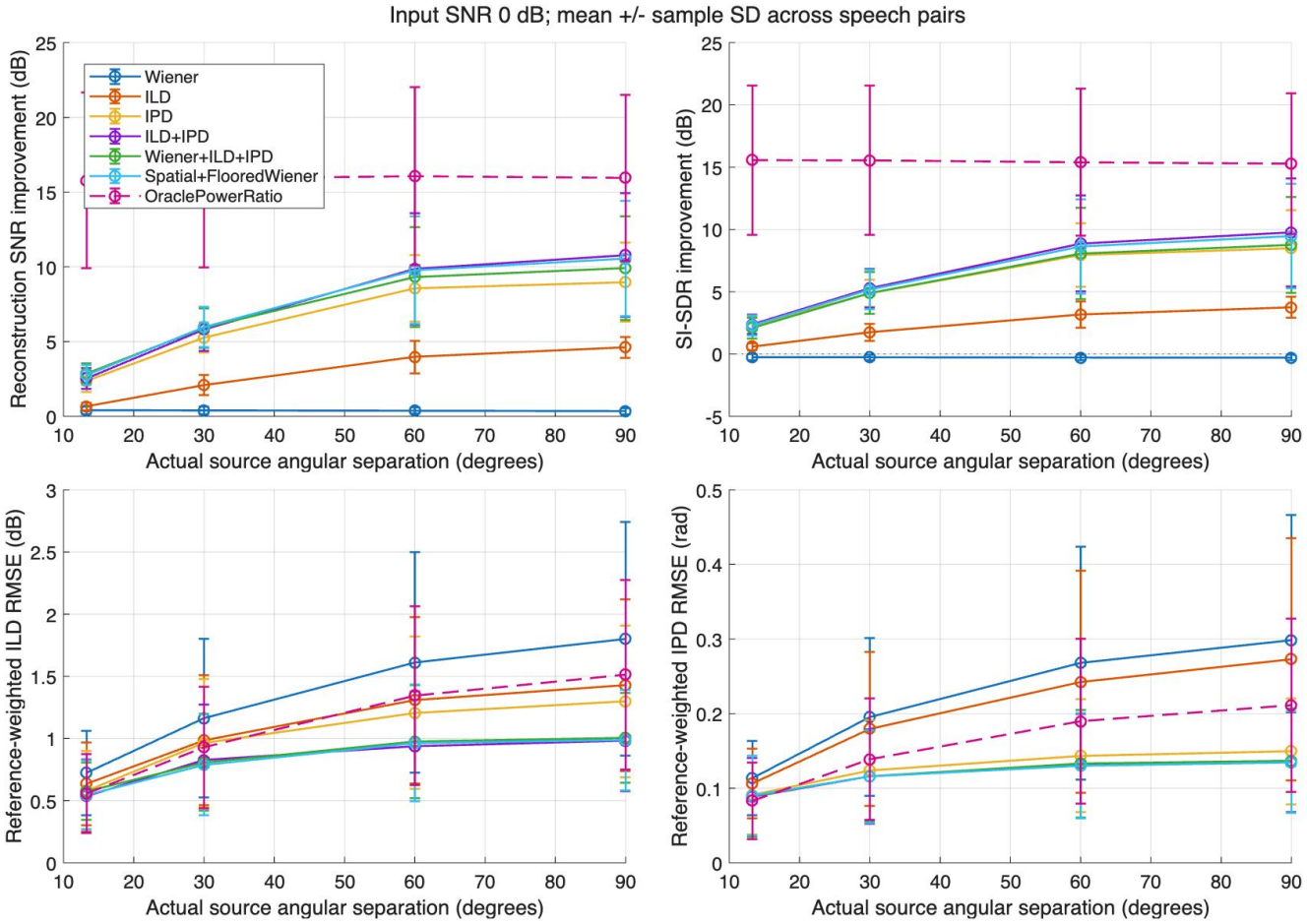


Figure 1. Performance at 0 dB input SNR.

Figure 1 shows the original MATLAB performance graph. The top panels show the reconstruction SNR gain and SI-SDR gain, and the bottom panels show the ILD/IPD error. Error bars are shown for the standard deviation of the samples from the three pairings. In the figure caption, ILD+IPD refers to the combined spatial mask, Wiener+ILD+IPD refers to the original product, and Spatial+FlooredWiener refers to the revised version. SI-SDR mean values are provided in Table 1. Better separation improves the spatial techniques, but Wiener alone is near zero improvement.

Table 1. SI-SDR improvement at 0 dB (dB).

Method	**13.3°**	**30°**	**60°**	**90°**
Wiener	-0.25	-0.26	-0.28	-0.29
ILD	0.60	1.75	3.18	3.75
IPD	2.28	4.88	7.95	8.49
Spatial	2.39	5.28	8.86	9.77
Original product	2.10	4.89	8.06	8.77
Floored product	2.30	5.17	8.63	9.47
Oracle	15.57	15.55	15.39	15.28

For 60˚ angle between speakers, masking based purely on ILD improves SI-SDR by 3.18 dB, whereas IPD-based masking results in an SI-SDR improvement of 7.95 dB and the combination achieves 8.86 dB. Clearly, the joint effect is better than either individual one, which means that level information can still be valuable even when the phase information is informative. This shows that there is complementary nature of cues in these scenes, not a universal ranking of human spatial cues. Phase mask uses a direction-specific HRIR template and its performance depends on the speech mixture and selected tolerance.

Joint spatial mean grows from 2.39 dB at 13.3˚ to 9.77 dB at 90˚. For small angular separations, the binaural cues in the two directions are more alike and therefore a target-compatible bin is less selective. When the separation is larger, the effectiveness of cue-based rejection increases. Joint spatial performance at 60˚ has the sample standard deviation of 3.85 dB and thus there is no guarantee that the angular gain will also give a similar gain for each individual utterance. Using only means would hide this variability. Oracle has the mean value of 15.39 dB at 60˚ angle and thus demonstrates that knowing the source powers might have yielded even greater improvements.

4.2 Effect of the Wiener floor

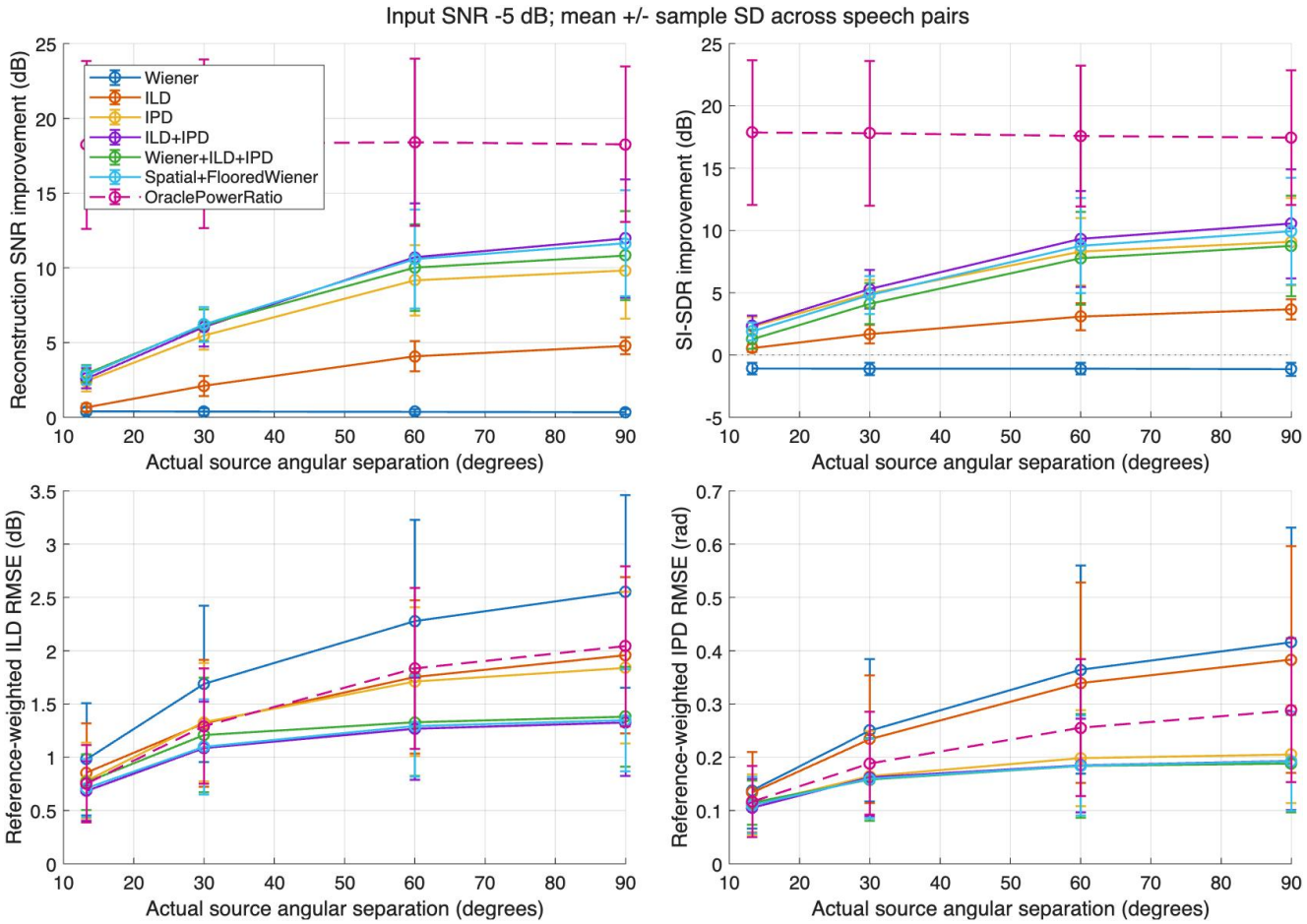


Figure 2. Performance at −5 dB input SNR.

The graph in Figure 2 shows the original performance plot using MATLAB with input SNR equal to -5 dB. In the upper-right corner of this figure, the original and floored product graphs as well as the spatial-only mask have been presented where stronger interference exists. Table 2 highlights all three SNR input values and shows the paired differences that result. For all 36 scenes, the revised performance improves the SI-SDR values for all the comparisons, resulting in an average gain of 0.553 dB, while the range extends between 0.001 and 1.883 dB. The reconstruction SNR value also improves by 0.295 dB.

Table 2. Mean performance across angles and pairs.

Input SNR	Spatial SI-SDRi	Original SI-SDRi	Floored SI-SDRi	Paired ΔSI-SDR	Paired ΔSNR
-5 dB	6.89	5.48	6.36	+0.876	+0.335

Input SNR	Spatial SI-SDRi	Original SI-SDRi	Floored SI-SDRi	Paired ΔSI-SDR	Paired ΔSNR
+0 dB	6.58	5.96	6.40	+0.440	+0.261
+5 dB	6.11	5.68	6.03	+0.343	+0.288

All values given in Table 2 are in units of decibels (dB). SI-SDRi is the difference with respect to the input signal mixture; Δ denotes the floored-product score minus the original-product score. The average paired SI-SDR gain of the method is 0.876 dB for -5 dB input, 0.440 dB for 0 dB, and 0.343 dB for +5 dB. This pattern agrees with the fact that the spectral factor causes more interference degrading at higher levels of interference presence. For -5 dB, the gain rises from 0.630 dB for 13.3˚ up to 1.171 dB for 90˚, because in this case, the spatial mask can keep the beneficial regions which are rejected by the spectral mask.

However, the comparison with the spatial-only masking approach is important as well. Its average SI-SDR gain is larger than that of the floored product at all investigated input signal-to-noise ratios: 6.89 dB against 6.36 dB for -5 dB, 6.58 dB against 6.40 dB for 0 dB, and 6.11 dB against 6.03 dB for +5 dB. Thus, although the modified approach improves the result with respect to the previous one, it does not exceed the performance of the spatial-only masking on this measure.

4.3 Suppression and target loss

Table 3. Passed source energy at 60° and 0 dB.

Method	Target retention	Interference retention	Output SIR	SNRi
Wiener	-0.61	-0.50	-0.11	0.37
IPD	-0.37	-9.32	8.95	8.56
Spatial	-0.58	-11.49	10.91	9.87
Original product	-1.17	-12.05	10.88	9.32
Floored product	-0.89	-11.79	10.90	9.77

Table 3 uses dB exclusively and separates source attenuation from the waveform score. The original product maintains the target at -1.17 dB, unlike the spatial-only masking at - 0.58 dB. The floor moves target preservation to -0.89 dB, gaining 0.28 dB of target energy compared to the original. In addition, interference retention increases from −12.05 dB to −11.79 dB. The reasons behind both effects are the increase in gain because the revised product cannot differentiate between the target and competing speech energy preserved as part of the mixed bin.

Signal-to-interference ratio (SIR) of the output is similar - 10.88 dB in the original product and 10.90 dB in the revision, but the improvement in SI-SDR increases from 8.06 to 8.63 dB. The combination of factors suggests that reducing target distortion should be considered the reason for gain instead of decreasing competing speech even further. Wiener-only masking keeps nearly identical portions of target and interference energies, which explains why the spectral attenuation shows low source discrimination.

A trade-off becomes evident at 13.3˚. The floor increases SI-SDR in all three input signal-to-noise ratios (SNRs), but reconstruction SNR reduces by about 0.10, 0.10, and 0.06 dB at -5, 0, and +5 dB of input SNR, respectively. At overlapping spatial patterns, increasing Wiener gain leads to competing energy transmission. SI-SDR can account for fitted gain of the target, while reconstruction SNR directly penalizes the amplitude error. Thus, the two metrics can favor different gain levels; an increase in SI-SDR cannot prove improvement of both metrics.

4.4 Binaural cue errors

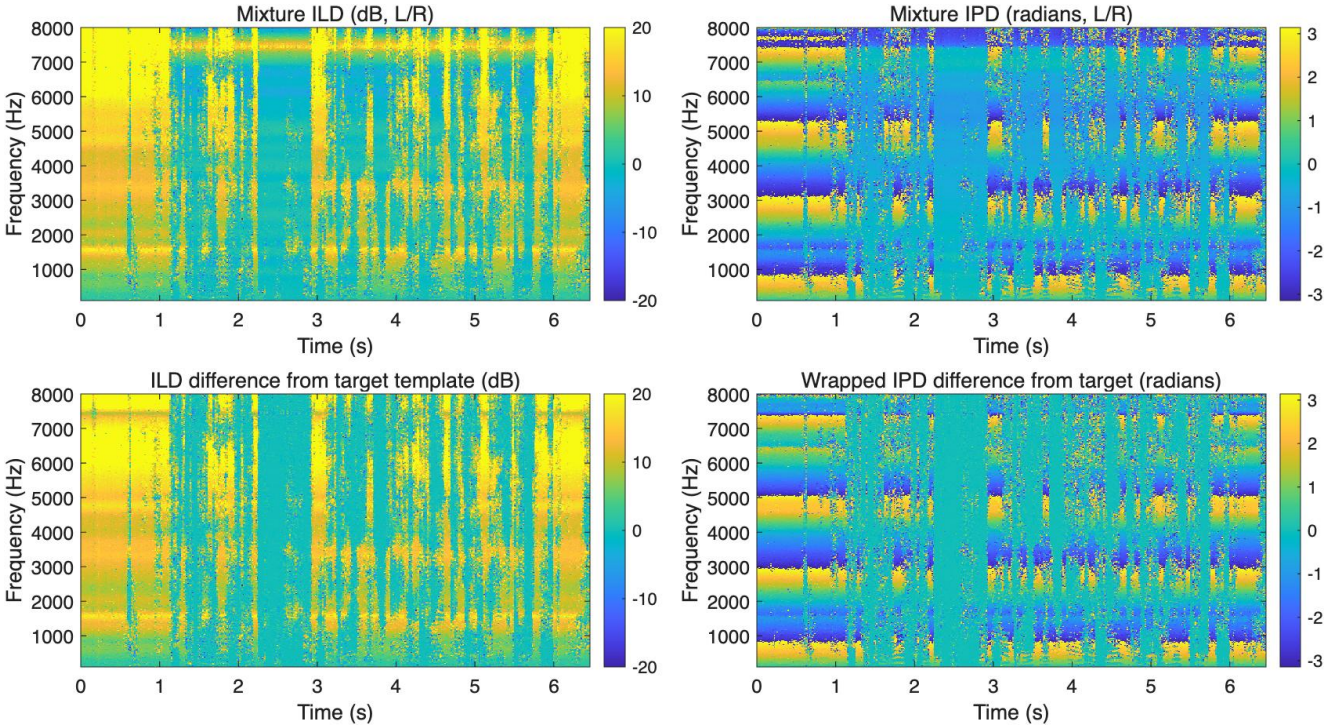


Figure 3. Mixture cues and target-template differences.

Figure 3 shows the original MATLAB cue maps for the demonstration scene, namely pair 1 at 60° and 0 dB. The top panels depict mixture ILD and IPD maps, whereas the bottom panels present the deviations from the target HRIR template in terms of ILD and IPD, phase wrapped to $\pm\pi$. Such maps act as input features rather than output error maps. It is seen that target compatibility is varying in time and frequency, making the local mask application preferable to assigning one gain for the entire utterance.

The bottom panels of Figure 1 depict spatial errors after reconstruction at 0 dB. Corresponding CSV results give mean ILD/IPD errors of 1.63 dB and 0.266 rad for the unprocessed mixture at 60°; such a reference is absent in Figure 1. Joint spatial estimation brings the errors down to 0.94 dB and 0.131 rad, whereas floored product gives 0.96 dB and 0.130 rad. Cue errors of the latter are similar to the spatial baseline and lower than those of the mixture; yet, slight differences between the fusion methods should not be taken as perceptual advantages.

Shared ear mask preserves the interaural ratio of the mixture in each modified STFT bin kept by the mask. Moreover, it cannot replace the ratio with the clean target one in bins with contributions from both sources. Finally, synthesis and reanalysis bring about STFT consistency effects. Therefore, nonzero reference errors are to be expected even with the shared mask. Greater source separation will increase source rejection in the wider source-separated areas but will leave greater cue differences in the rest of the overlap region; thus, spatial RMSE need not follow the angular trend of SI-SDR.

The minimum cue-energy coverage among all scored outputs is 0.9967. Hence, low errors are not due to discarding most target-reference energy as undefined output cues. However, the scores are weighted by target energy and do not represent perceptual direction or the whole residual sound scene. Low- and high-band CSV diagnostics are provided for viewing; lacking a frequency-band masking experiment, they cannot prove a new frequency assignment rule for ILD and IPD.

4.5 Time–frequency inspection

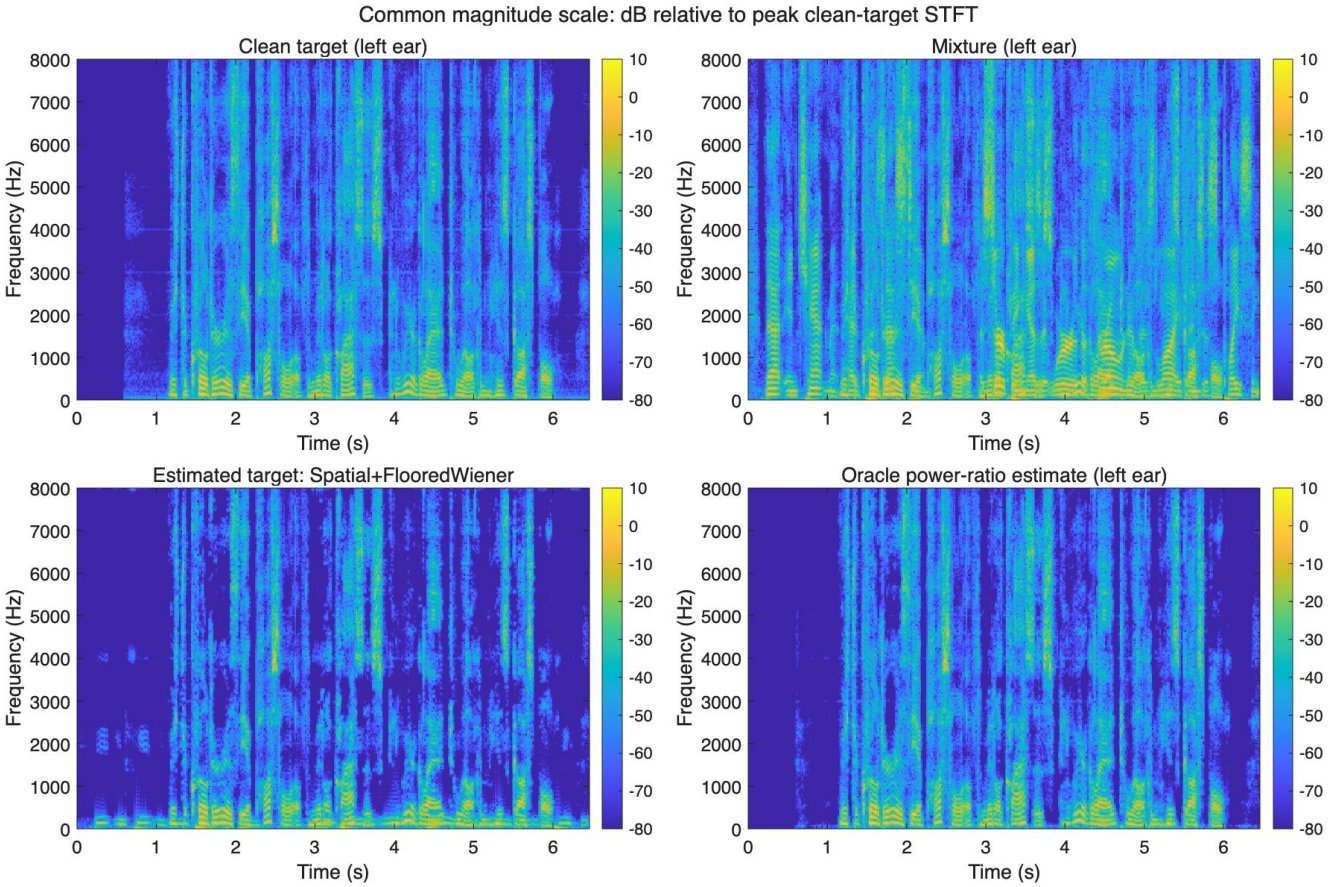


Figure 4. Left-ear spectra for the demonstration scene.

Figure 4 shows the original MATLAB spectrogram comparison for pair 1 at 60˚ and 0 dB. The graphs show the clean target, the mixture, the floored-product estimate, and the oracle power-ratio estimate. In all four graphs, the scale is from -80 to +10 dB relative to the clean-target STFT peak. Warmer colors indicate more energy. The clean-target reference shows the correct speech structure that needs to be preserved while the mixture includes extra interfering speech. Oracle provides a known-component comparison instead of an actual output.

The graph makes it easier to find the lost targets and the residual energy; however, in this case, it is impossible to assign the overlapping sources in both time and frequency domains to one specific speaker automatically. High output energy may indicate either the target information recovered or the interfering noise added. The passed-component measure in Table 3 helps to find out which of the two cases is true. The spectrograms help analyze the numerical balance but do not indicate intelligibility, musical noise level, or the listener preference. Eighteen common gain WAV files are provided for further tests.

4.6 Spatial re-mixing

Table 4. Re-mixing with matched clean references.

Component	SNR (dB)	SI-SDR (dB)	ILD RMSE (dB)	IPD RMSE (rad)
Target	6.40	5.30	0.228	0.035
Residual background	4.71	2.44	0.203	0.024
Combined	12.50	12.32	2.481	0.433

In particular, the SI-SDR of the isolated re-rendered source is 5.30 dB, while cue errors are 0.228 dB for ILD and 0.035 rad for IPD. For the residue part, SI-SDR equals 2.44 dB. Meanwhile, the SI-SDR of the whole scene equals 12.32 dB; however, the error values are higher, i.e., 2.481 dB and 0.433 rad for ILD and IPD, respectively. These numbers are based on different baselines and hence cannot be compared directly as measures of separation quality. In particular, the cue information related to the isolated parts is determined by the new HRIR; cue information of the scene depends on levels and leakage between both parts.

5 Discussion

The main finding is that known direction spatial compatibility provides discrimination capacity within these mixtures. When there are larger angles, the target and interference have sufficiently distinct measured binaural signatures such that the combination mask is better than each mask alone and the basic Wiener filter baseline. This finding is consistent with cue overlap considerations discussed in references [1,2]. However, published MESSL and neural scores cannot provide a numerical baseline for comparison as these methods have not been run on this dataset with matching references.

This fusion experiment highlights a particular flaw of the approach. Multiplying two attenuating masks can excessively suppress target energy even when one mask considers the bin compatible with the target. The gain floor reduces this extra spectral filtering and improves all matched SI-SDR results. Target preservation and output SIR calculations show that the effect is primarily due to improved attenuation balancing and not better discrimination of the sources. Spatial only masking provides the best SI-SDR on average among practical variations.

Finally, these results explain the need for evaluation from multiple perspectives on the output. A gain configuration which improves SI-SDR may reduce reconstruction SNR for similar directions, and a low error on single component cues after re-mixing mostly indicates HRIR filtering. Binaural enhancement literature takes scene preservation into account along with suppression [3,4]. Here, waveform scores, passed source energies and spatial errors tackle different aspects of the question; none alone defines listening quality.

Therefore, a sensible spatial configuration of the newly rendered components cannot be considered proof of the source extraction task completion. Fold-down can produce cancellation and retains the previous HRIR spectral coloration. Table 4 analyzes the errors that emerge during an explicitly matched processing procedure, rather than the accuracy of the localization or faithfulness to the dry source. A listening test should be performed to establish if the scene is easier to understand or localizes correctly.

The data is limited to only three utterance pairs of different lengths, using only one HRIR listener, static sources and a silent target calibration. The same speech-based diagnostics were used in the evaluation of the original product, therefore the additional SNR values do not correspond to the validation of the source. The 36 conditions are interdependent and there are no results in terms of statistical significance and generalization to realistic audio recordings. Directional estimation, room reverberation, dynamic sources, neural adaptation and subjective experiments have not been included in the experiment.

For a more reliable follow-up experiment the tolerance and the fusion weight will need to be selected based on a separate pair of development examples, and restricted to novel utterances and HRIR listeners. Evaluating the calibration failure, directional misalignment and room reverberation will help to determine the limitations of the technique. A matched executable version of MESSL or binaural neural baseline algorithm will show how much the fixed cues limit the performance. Listening tests in headphones with the ratings will estimate the speech intelligibility, artefacts and localization.

6 Conclusion

The results obtained from measuring ILD and IPD lead to better source separation relative to the baseline of spectral separation in the task performed, with higher mean improvements being noted for larger angular separations. For a separation of 60° and 0 dB, the SI-SDR improvement scores for the joint spatial, original product, and floored product are 8.86, 8.06, and 8.63 dB, respectively. The lower-bounded Wiener filter improves upon its ancestor by a mean of 0.553 dB for 36 similar scenes, but the spatial-only mask still proves to be a superior method, on average. All in all, this research highlights the functioning of a binaural separation and re-mixing process and a measured improvement whose strengths and weaknesses can be linked to target preservation and interference.

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